

Shreyas Dhekane

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Education

Indiana University, Bloomington

Master of Science, Computer Science

• GPA: 3.45/4

• Coursework: Object-Oriented Software Development, Applied Algorithms, Cloud Computing

Aug 2025 - May 2027

Bloomington, IN

D. Y. Patil Deemed to be University

B.Tech, Computer Science and Business Systems

• GPA: 8.76/10

Aug 2021 - May 2025

Mumbai, MH

Experience

fastflo

Software Engineer Intern

- Built, tested, and shipped reliable full-stack features for the Fastflo platform using TypeScript, HTML5, SQL, and WebSockets, improving platform maintainability and software quality.
- Built a global command palette enabling instant navigation, quick filters, and candidate/position search across the platform, reducing navigation time by 65%
- Built and maintained RESTful APIs and integrations with Entity Framework and SQL that support automated, multi-channel candidate messaging, improving message delivery reliability
- Translated product requirements into production-ready software by partnering with engineering and product teams, accelerating delivery through Agile development and TypeScript.

Jun 2026 - Aug 2026

Cambridge, MA (Remote)

Quetware Technology

Web Intern

- Designed and improved responsive, component-based UIs in Next.js and TypeScript, collaborating with engineers to connect front-end and back-end services, which sped up feature launches and reduced user-reported issues.
- Reduced design-to-dev handoff time by 20% through reusable components and Git-based workflows, improving CI/CD deployment speed.
- Enhanced user engagement through usability testing and iterative prototyping, boosting average session duration by 18% and reducing bounce rate by 12%

Jun 2024 - Aug 2024

Remote

Projects

Ryde Full-Stack Ride-Hailing Mobile App

- Expo, React Native, TypeScript, Node.js, Express, Prisma, PostgreSQL, Socket.IO
- Architected and validated the data layer across 12 Prisma models and 11 relations, exposed through 20 REST endpoints (16 authenticated) powering a 15-screen Expo/React Native TypeScript client.
- Built and tested a real-time trip pipeline on Socket.IO with 8 distinct event types driving live driver matching, location updates, and in-trip chat across 3 mobile screens.
- Established a shared design system of 5 token groups consumed across 20 files, keeping 39 TypeScript interfaces consistent app-wide, and cut image payload 97%.
- Built the fare and payments layer saved-card CRUD, promo-code discount logic, and 2 additional fare tiers, while collapsing the booking flow steps by 50%.

Aug 2026

Kraken2D Custom 2D Arcade Engine in C++

- C++17, SDL3, CMake, Fixed-Timestep Physics, Procedural Audio
- Re-architected engine from an Entity Component System to a flat header-per-game design after profiling revealed ECS overhead dominated runtime at sub-200 entity scale, reducing engine core by 62%.
- Built a fixed-timestep game loop using an accumulator pattern to decouple physics from rendering, eliminating tunneling issues and reducing frame-time variance by ~85% across variable frame rates without external libraries.
- Developed a procedural audio engine generating sine, square, and noise waveforms in real time via SDL3 audio streams with ADSR envelope control, shipped with zero audio assets, eliminating asset pipeline and licensing overhead.
- Implemented impulse-based AABB and circle collision resolution with positional correction; evaluated SAT and GJK but selected minimal primitives based on actual gameplay needs, reducing collision code by ~40% without sacrificing correctness.

May 2026

Pulsar Real-Time Distributed API Monitoring Platform

- C#, .NET, ASP.NET Core, SignalR, PostgreSQL, Azure App Service
- Designed RESTful APIs and real-time communication using WebSockets (SignalR) to stream live system health updates, improving connection reliability by 60% through automatic reconnection and long-polling fallback.
- Computed p50/p95/p99 response time percentiles across a live Angular dashboard, surfacing tail latency spikes that average-based monitoring missed in 30% of degradation events.
- Reduced false positive alerts by 45% by anchoring anomaly detection to a rolling 24-hour baseline, and integrated Google Gemini API to generate natural language health summaries from raw metrics.
- Deployed and configured the backend on Azure App Service with a managed Supabase PostgreSQL database, using environment scoped secrets for JWT authentication and third-party API keys.

Feb 2026

Technical Skills

- Languages: C++, C#, JavaScript (ES6+), TypeScript, Python, Java, SQL
- Systems & Backend: ECS Architecture, Fixed-Timestep Physics, Collision Detection, ASP.NET Core, .NET, SignalR, REST APIs, Entity Framework, PostgreSQL, Supabase
- Frontend: Angular, React, Next.js, HTML5, CSS3, Tailwind CSS
- Concepts: Data-Oriented Design, Real-Time Systems, WebSockets, API Design, Cloud Deployment, Agile/Scrum, Docker
- Tools & Platforms: Git, GitHub, CMake, SDL3, Vercel, AWS, CI/CD, Azure App Service